LOW-FIDELITY & HIGH-FIDELITY

Prototyping in Human–Computer Interaction (HCI)



Learning Objectives

1. Objective

Understand the role of prototypes in HCI

2. Objective

Differentiate low-fi and high-fi prototypes

3. Objective

Compare their advantages and limitations

4. Objective

Learn where each prototype is used in the design process

5. Objective

Apply the concept to real UI/UX projects

HCI and the Role of Prototyping

Why do designers create prototypes?

HUMAN

Users, needs, behavior, expectations and usability

INTERACTION

How users navigate, click, understand and complete tasks

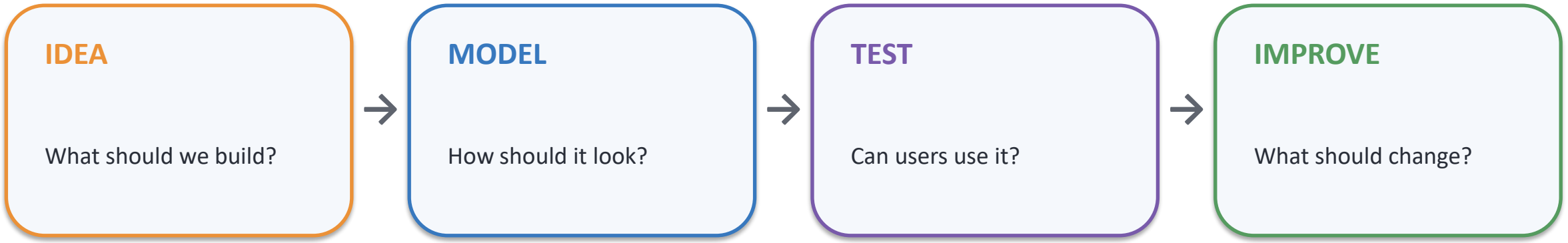
COMPUTER

Interfaces, applications, websites and digital products

HCI Goal: Build systems that are useful, usable, efficient and satisfying.

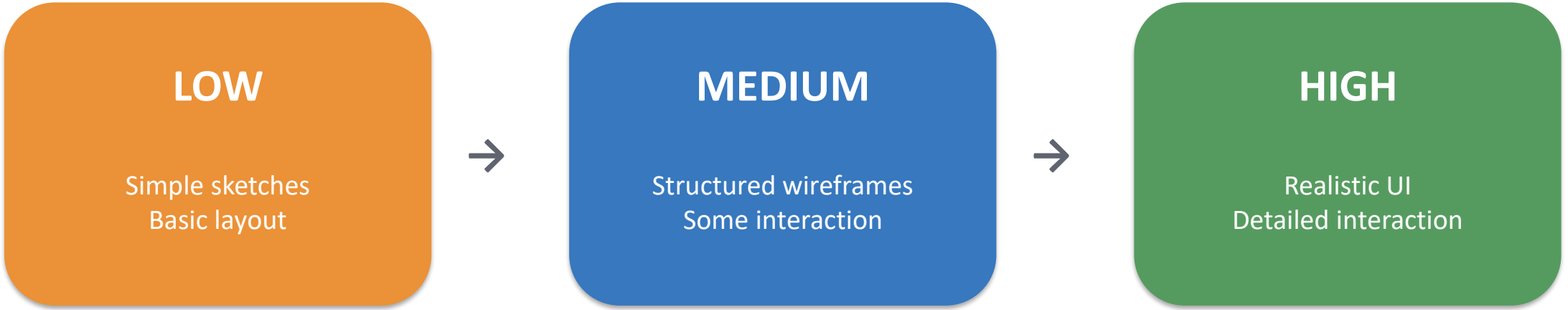
What is a Prototype?

A prototype is an early model of a product that allows designers and users to explore and test ideas before the final system is developed.



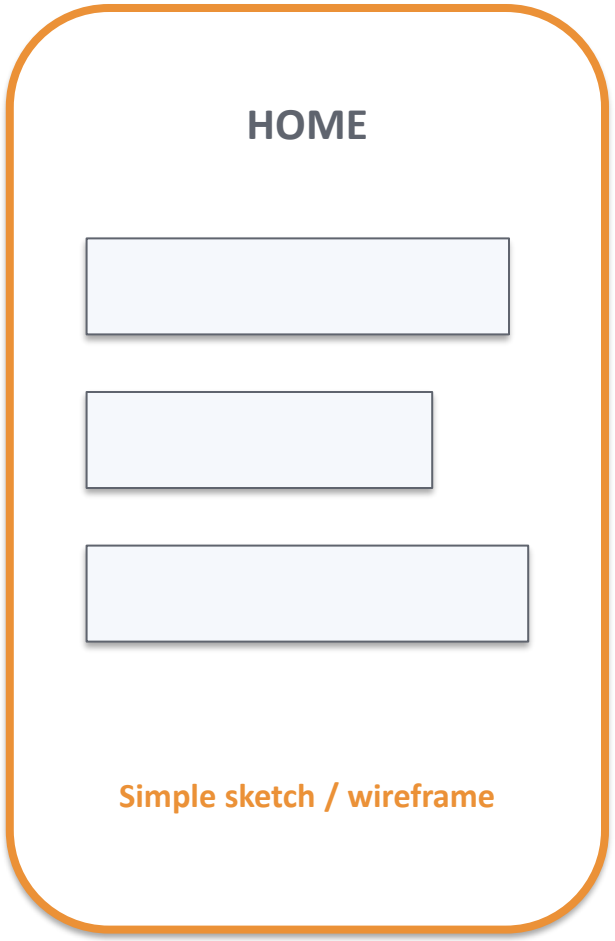
Understanding Fidelity

Fidelity = how closely the prototype resembles the final product



As fidelity increases, realism, detail, interaction, time and cost usually increase.

Low-Fidelity (Low-Fi) Prototype



Definition

A simple representation focused on layout, structure, navigation and ideas.

Paper sketches

Whiteboard designs

Wireframes

Simple clickable screens

Low-Fi: Advantages and Limitations

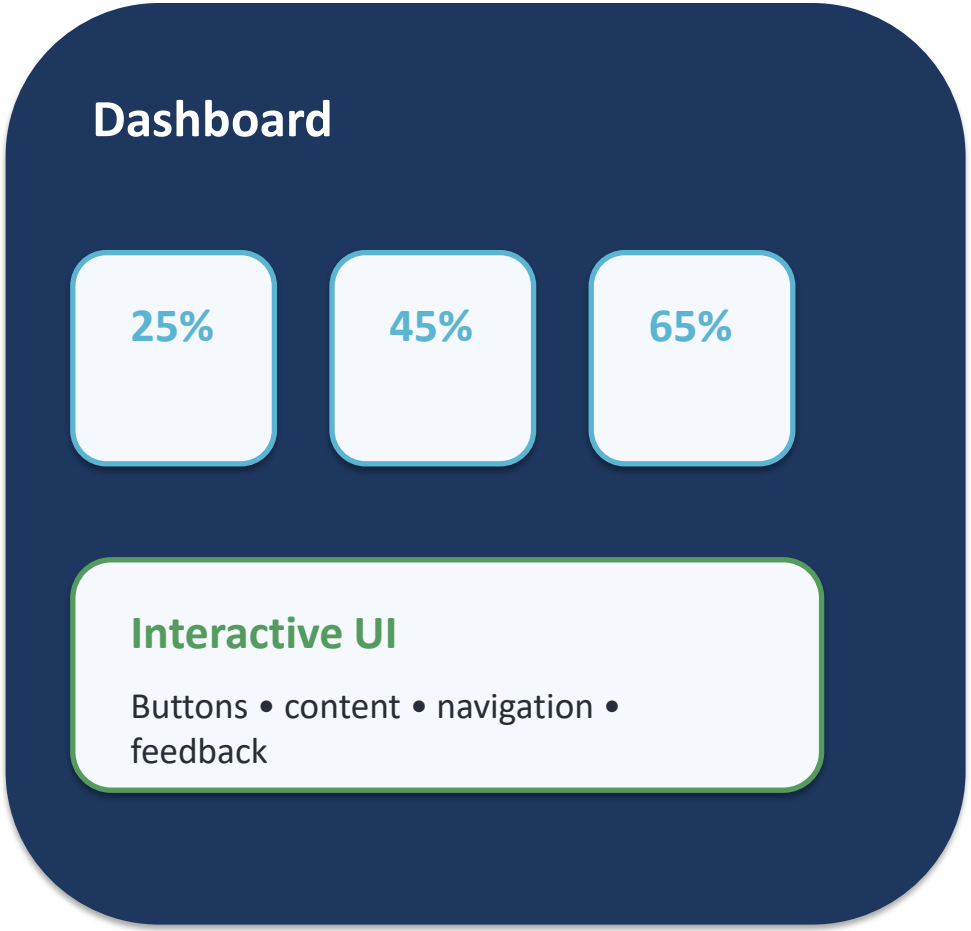
✓ ADVANTAGES

- Fast and inexpensive
- Easy to modify
- Encourages brainstorming
- Supports early feedback
- Focuses on functionality
- Reduces development risk

⚠ LIMITATIONS

- Not visually realistic
- Limited interactions
- Difficult to test animations
- Users may struggle to imagine final UI
- Not ideal for final client demonstrations

High-Fidelity (High-Fi) Prototype



Definition

A realistic and detailed prototype that closely resembles the final product.

Real colors & typography

Images & icons

Clickable interactions

Animations & transitions

High-Fi: Advantages and Limitations

✓ ADVANTAGES

- Realistic user experience
- Detailed usability testing
- Strong stakeholder presentations
- Tests visual hierarchy
- Tests interaction and workflow
- Supports developer handoff

⚠ LIMITATIONS

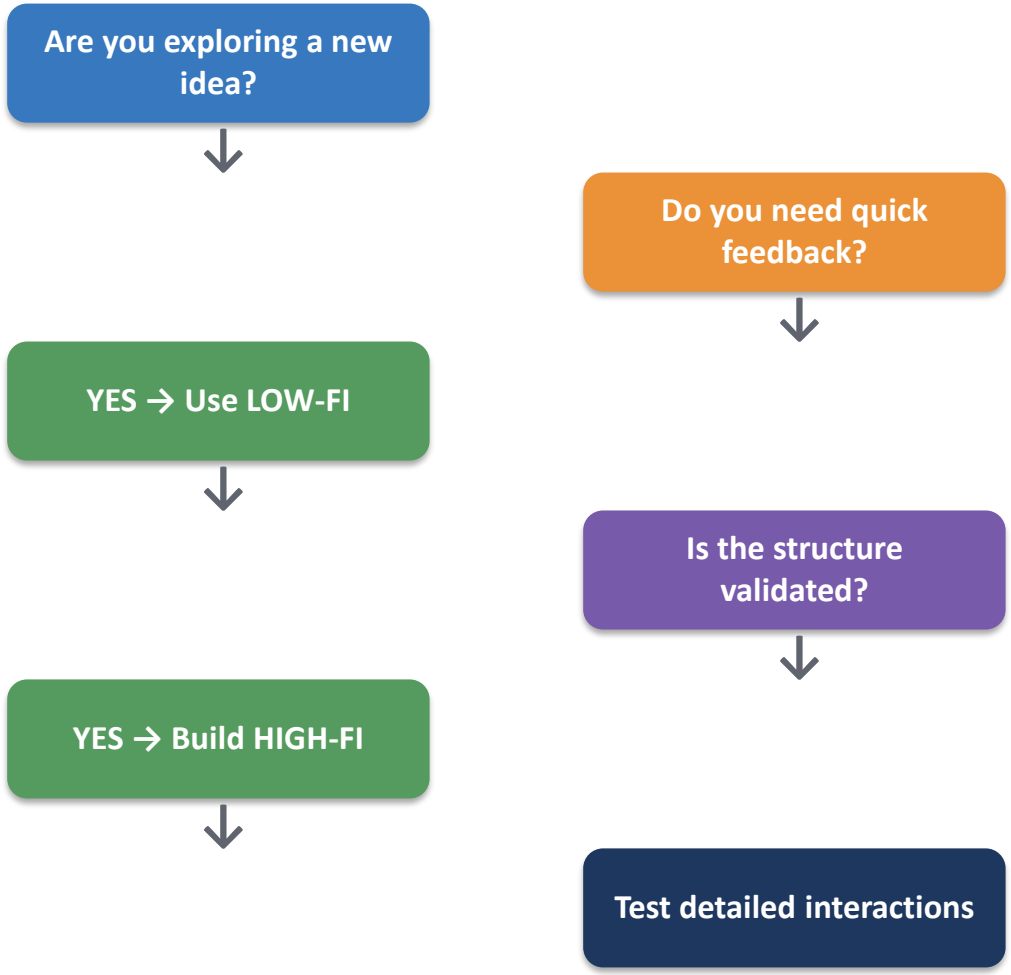
- More time-consuming
- Higher cost
- Requires design skills
- Changes may take longer
- Users may assume product is finished

Low-Fi vs High-Fi: Detailed Comparison

Feature	LOW-FIDELITY	HIGH-FIDELITY
Appearance	Rough and simple	Detailed and realistic
Cost	Low	Higher
Creation Time	Minutes to hours	Hours to days/weeks
Interactivity	Limited	Realistic and detailed
Best Stage	Early design	Later design
Main Focus	Ideas and structure	Experience and interaction
Examples	Paper sketch, wireframe	Interactive Figma UI
Changes	Very easy	More time-consuming

Which Prototype Should You Use?

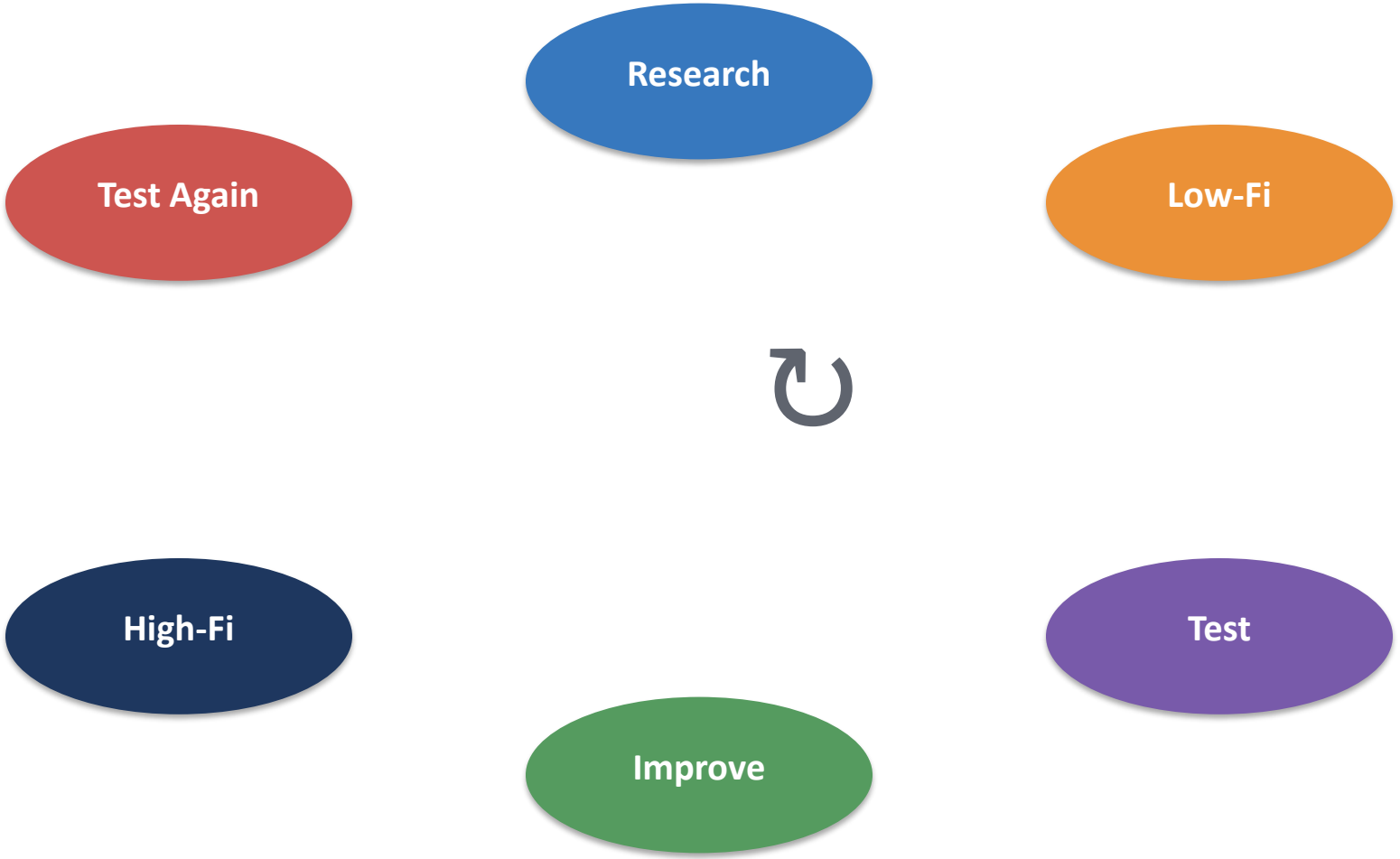
Decision Flowchart



Rule of thumb: Start simple → learn from users → increase realism only when needed.

HCI Prototyping Cycle

An iterative design process



Iterate until usability goals are achieved

Role of Prototypes in Usability Testing

LOW-FI TESTING

Navigation
Information structure
Missing features

USER FEEDBACK

Observe users
Ask questions
Identify confusion

HIGH-FI TESTING

Visual hierarchy
Interactions
Detailed workflow

Goal: Find usability problems before spending resources on full development.

Visual Example: Food Delivery App

LOW-FI VERSION

[Logo]

[Search Bar]

[Restaurant Box]
[Restaurant Box]

[Cart Button]

→

HIGH-FI VERSION

🍔 Food images and realistic content

🔍 Styled search and filters

⭐ Ratings and prices

🛒 Interactive cart

📱 Clickable navigation and checkout

Same idea — different levels of detail and realism.

Common Prototyping Tools

LOW-FI TOOLS

Paper & Pen
Whiteboard
Balsamiq
Basic Wireframes

HIGH-FI TOOLS

Figma
Adobe XD
Sketch
Framer / Axure

Choose the tool based on the design stage—not simply because it has more features.

Class Activity

STEP 1

Choose an app:
University Portal / Shopping / Food Delivery

STEP 2

Draw 3 low-fi screens:
Home → Details → Action

STEP 3

Convert into high-fi:
Colors, images, icons, interactions

STEP 4

Test with a classmate and collect feedback

Summary

LOW-FI

Fast • Cheap • Flexible
Explore ideas early

→

HIGH-FI

Realistic • Interactive • Detailed
Test the complete experience

KEY MESSAGE

Start simple, test early, improve continuously, and increase fidelity when necessary.

THANK YOU

Questions & Discussion

Human–Computer Interaction (HCI)